

Evaluating Non-Generative Decision Models in Sequential Gridworld Tasks

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Abstract

Non-generative decision models assign probabilities to supplied options without writing an answer. System-One Control Bench tests whether these decisions can guide navigation across 100 symbolic gridworld puzzles, ten input conditions and five action spaces. An exact solver identifies every optimal move, including ties. DeepSeek V4.1 Flash (reasoning) has the highest won rate among the models: 80% with full context. Jev wins 57%; paired comparisons do not establish an advantage over Gemma or DeepSeek without reasoning. Supplying simulated outcomes improves Jev’s navigation, but larger sets of sequence moves do not consistently help. Laya and GLiClass stay near random with the map alone. On shared exam positions, Jev and Gemma both reach 90% full-context optimality. For Jev, this exceeds its own-game optimality even after matching puzzle weighting: good isolated decisions do not ensure completed games.

1 Introduction

A non-generative decision model returns probabilities over supplied options. For sequential control, those selections must compose into a route, rather than merely look plausible individually. Jev’s use in chess and software control motivates testing this distinction in an inspectable environment [1], [2], [3]. We study a fully observed symbolic world: the map already contains the information needed to navigate, while the controller varies its representation and the execution between observations.

“System-One” refers to fast decisions without deliberate reasoning, after the System 1 / System 2 distinction; Laya calls its decision interface `system_one`.

Four research questions guide the report:

1. **RQ1:** How well do non-generative models complete navigation from a complete symbolic state?
2. **RQ2:** How do local descriptions, interaction history, simulated action outcomes and explicit subgoals affect performance?
3. **RQ3:** How does performance vary across action formulations with different sequence lengths, option counts and observation intervals?
4. **RQ4:** How informative are returned probabilities, and what performance is obtained at the measured inference cost and latency?

We contribute a controlled environment with exact, tie-preserving move labels; a diagnostic evaluation of representations and action formulations; and empirical results with raw trajectories, replay validation and reproducible analysis. Solver verification is established in prior planning evaluation [4]; our focus is its diagnostic use for non-generative decision models.

2 Related work

Jev has been evaluated as a legal-move selector in chess [1] and compared with Laya and a local Qwen3 decision model [5]. Laya’s evaluated checkpoint specializes in four workflow domains and cautions about transfer [6]. GLiClass performs zero-shot classification over supplied labels in a single forward pass [7]; treating option texts as labels lets us evaluate such a classifier as a sequential controller.

PlanBench tests planning and reasoning about change in formal domains [4]. AgentBench covers eight interactive environments [8], and BALROG evaluates agents across games of differing difficulty [9]. We instead vary input components within one small environment to diagnose non-generative

decision models. BabyAI [10] and MiniGrid [11] provide related gridworld tasks; compass moves here separate map interpretation from orientation control.

SayCan combines language-based skill selection with feasibility estimates [12], while SayCanPay adds learned domain knowledge [13]. Our simulated outcomes are deterministic transitions supplied by environment code. Multiple-choice selection is sensitive to option identifiers and order [14], motivating seeded shuffling without assuming it removes bias. Calibration methods distinguish confidence from correctness [15]; our target accepts any solver-optimal move, rather than a unique class label.

3 Benchmark design

A puzzle contains walls (#), floor (.), a moving piece (A), goal (G), and optionally a key (K) and locked door (D). Compass steps move north, south, east or west. Collecting the key allows passage through the door; a wall or locked door otherwise blocks movement. Reaching the goal wins. Breadth-first search over position, inventory and door state labels every move that begins a shortest route, preserving ties.

```
0123456789
0 #####
1 #.....AX#
2 #..##*K.#
3 #D***#..#
4 #G#.#...##
5 #####
```

Map alone: position (7, 1), empty inventory, map and object coordinates; “What is the best next move?”

Simulated outcomes: “move south (down): you move to (7, 2) and pick up the key”. Shortest route: south, west, west, south, west, west, west, west, south.

Figure 1: The frozen example position. Stars trace one shortest route through K and D to G; X marks east, a suboptimal move toward open floor. Stars and X replace floor only in this figure. Appendix A gives its full-context request.

The 100 puzzles span levels 1, 2, 3, 4, 5, 6, 8, 10, 12, 15 and 20, where level is shortest distance in compass steps. Levels 1 and 2 contain five puzzles each; others ten. Seeded generation creates rooms and mazes, with one hand-made level-10 maze. Higher levels add mandatory detours and key-door dependencies; the wall-aware greedy baseline fails on all 40 puzzles at levels 10 and above. Extra outer-wall rings in half of the puzzles at levels 12, 15 and 20 add irrelevant map text without changing the interior.

A **move** is one model decision; a **step** is a compass displacement within it. Action spaces offer one step (compass, 4 options), exactly two (two-moves, 16), up to two (up-to-two-moves, 20), exactly three (three-moves, 64), or up to three (up-to-three-moves, 84). All sequences are offered. A blocked step is wasted and execution continues; reaching the goal immediately ends the move. The next observation follows execution. Changing the formulation changes option count, text length and steps executed before the next observation; several sequences can end at the same position. These experiments compare the whole interface rather than option count alone.

Every request includes rules, the numbered map, position, inventory and object coordinates. A **local state description** explains adjacent cells and relative object locations. **Interaction history** lists prior moves and their results. **Simulated action outcomes** tell the model what would happen after each option. An **explicit subgoal** names the next target. The local descriptions and simulations are computed from information already present in the map. History is not needed to reconstruct the state, but may expose loops. The ten conditions add each component to the map alone or remove it from full context (Table 1).

Table 1: Condition-ID mapping. Full context includes all four components.

Artifact ID	Paper name
map	Map alone
map+surroundings	Map with local state description
map+memory	Map with interaction history
map+lookahead	Map with simulated action outcomes
map+subgoal	Map with explicit subgoal
everything	Full context
everything-surroundings	Full context without local state description
everything-memory	Full context without interaction history
everything-lookahead	Full context without simulated action outcomes
everything-subgoal	Full context without explicit subgoal

4 Evaluation protocol

4.1 Models and baselines

Random chooses uniformly from the offered moves. Greedy moves toward the next target, and greedy-walls also avoids walls. The solver always chooses an optimal move and provides a ceiling. The greedy baselines use compass moves only. Each model receives the request without direct access to the environment or solver.

Jev (`jev-1.13.0`) chooses the option with the highest returned probability. GLiClass (`modern-large-v3.0`) treats the state as classification text, the question as the task, and the option texts as labels. Its multi-label pipeline scores each option independently from 0 to 1; the controller divides these scores by their sum. Qwen3.5-4B does not write an answer: the controller scores each option’s number in its chat template, renormalizes over allowed option numbers, and selects the most likely one.

Laya receives the full state as the first argument of `system_one`, together with the question and option texts. Only the option IDs are omitted, because Laya would print them within each option’s 48-token allowance. The controller expands the token limits to fit the full request. It rejects an option above 48 tokens or a request above 8,192 tokens instead of truncating it. The evaluated checkpoint was trained for four workflow domains and warns about use outside them and large option sets [6]. Our results concern this checkpoint’s transfer to navigation under these input limits.

Gemma and DeepSeek use OpenRouter with DeepInfra as the fixed host. They return an option ID through a JSON schema. Direct answers have a 64-token limit. DeepSeek V4.1 Flash (reasoning), artifact ID `deepseek-v4.1-flash-think`, allows 1,024 reasoning tokens plus 64 answer tokens and is evaluated under compass moves. OpenRouter models retry refused calls four times, waiting 5 to 60 seconds; the reasoning timeout is 600 seconds. Jev retries refused calls twice but does not retry a 120-second timeout. A missing, invalid or capped answer ends the game as an error and counts as a loss. All 242 evaluated model/rules/condition cells have zero games ended by an error. Option order is shuffled using a seed based on the puzzle and move number. Each saved game provides one trajectory. Prompt development preceded puzzle generation; Appendix A.1 gives the chronology and later question correction; Appendix C records the inference setup.

4.2 Metrics and statistics

Won rate is the share of games reaching the goal within the move limit, twice the solver’s fewest moves. Progress is one minus the closest compass-step distance reached divided by the starting distance; final-state progress uses the ending position and can be negative. SPL is zero for a loss; for a win, it divides the fewest moves by the larger of that number and the moves used [16]. SPL and the move limit count model decisions. Optimality minimizes decisions under the selected rules and accepts every tied best move, including sequences with wasted steps. A blocked move leaves the board unchanged.

The frozen fixed-length subgoal prompts say “shortest path”; up-to prompts say “the fewest turns.” Evaluation uses the decision-count objective for both. Rounding to whole decisions also changes the attempted-step budget: a four-step puzzle permits four three-step moves (up to twelve steps), versus eight compass moves. Neither the wording nor this budget effect is controlled separately.

Intervals are 95% percentile-bootstrap intervals over puzzles, with 10,000 resamples. Component comparisons pair the same puzzles. McNemar’s exact test compares whether each game was won, and Holm correction covers the eight component contrasts for each model. Calibration pools all ten compass conditions. Ten-bin ECE and binary Brier score compare the chosen-option probability with whether that move was optimal [15]. Accepting several tied best moves is different from validating a probability distribution over alternative moves. Every evaluated model/input has 100 games.

The fixed-state exam uses 495 compass positions from 100 puzzles, with seed 0: five per puzzle, or four at level 1. Item counts are 100 start, 95 on-route, 144 off-route, 88 after-blocked and 68 late-route (after the key where present); missing kinds are filled with labelled off-route or after-blocked items, sometimes reusing a board. Scripted prefixes use the ordinary game, preserving history and option shuffles. Each model answers once per item under map alone and full context

Table 2: Compass won rate and efficiency. Each model/input has 100 games; intervals resample puzzles; baselines use the map alone. Full means full context.

Model/baseline	Map: won [95% CI]	Full: won [95% CI]	Map: progress / SPL	Full: progress / SPL
Solver	100% [100, 100]	—	1.000 / 1.000	—
Greedy (walls)	37% [28, 46]	—	0.475 / 0.370	—
Greedy	28% [20, 37]	—	0.356 / 0.280	—
Random	4% [1, 8]	—	0.249 / 0.033	—
Jev	32% [23, 41]	57% [47, 66]	0.425 / 0.319	0.683 / 0.541
Laya	2% [0, 5]	12% [6, 19]	0.105 / 0.020	0.264 / 0.105
GLiClass	3% [0, 7]	13% [7, 20]	0.104 / 0.030	0.216 / 0.130
Gemma 4 26B	35% [26, 44]	61% [51, 70]	0.452 / 0.322	0.714 / 0.568
Qwen3.5-4B	14% [8, 21]	45% [35, 55]	0.335 / 0.124	0.661 / 0.387
DeepSeek V4.1 Flash	39% [30, 49]	60% [50, 69]	0.545 / 0.347	0.715 / 0.542
DeepSeek V4.1 Flash (reasoning)	67% [57, 76]	80% [72, 87]	0.787 / 0.618	0.864 / 0.741

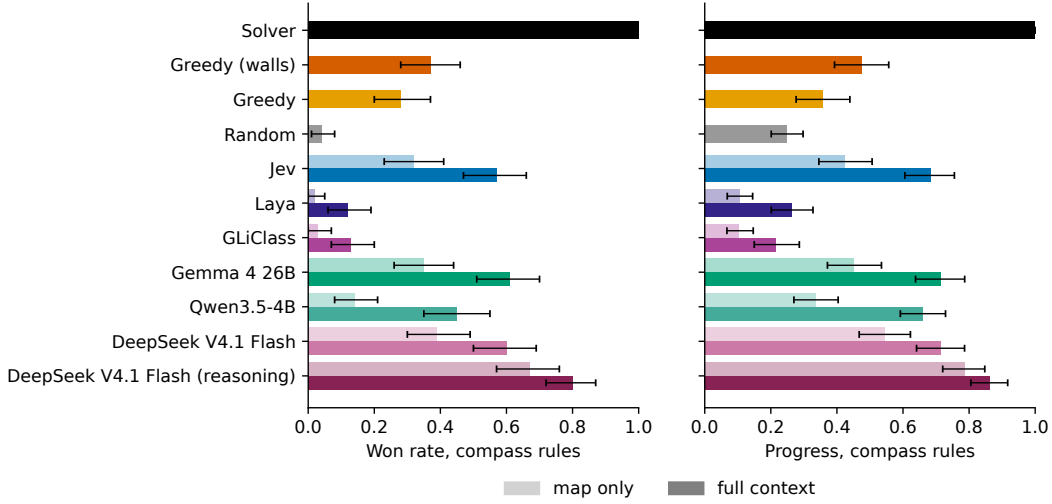


Figure 2: Compass won rate and progress with the map alone (pale) and full context (darker). Intervals resample puzzles. Each model/input uses 100 games.

(990 answers). Errors are incorrect; any solver-best move is optimal. Rates average within puzzles, then over puzzles, with intervals resampling puzzles. Random selects a fixed shuffled slot; chance optimality is 0.27.

5 Results

5.1 Won rate and efficiency

DeepSeek V4.1 Flash (reasoning) wins more games than any other model, with non-overlapping intervals under both inputs. Its won rate is 80% [72, 87] with full context and 67% [57, 76] with the map alone. Each estimate uses 100 games.

Jev wins 32% with the map alone, below the wall-aware greedy baseline’s observed 37% and above random’s 4%. With full context, Jev, Gemma and DeepSeek without reasoning win 57%, 61% and 60%. Paired won-rate differences for Jev minus Gemma are -3 percentage points $[-11, 5]$ with the map and -4 $[-10, 2]$ with full context; against DeepSeek they are -7 $[-14, 0]$ and -3 $[-11, 5]$. These intervals establish no advantage for Jev, without showing equality or inferiority. Laya and GLiClass win 2% and 3% with the map, near random; full context raises them to 12% and 13%.

5.2 Effects of input components

We compare the same puzzles with and without each component. For Jev, adding simulated outcomes to the map raises won rate by 16 percentage points [9, 23] and progress by 0.18 [0.13, 0.24]. Adding interaction history raises progress by 0.15 [0.11, 0.21]. Both additions pass the Holm-

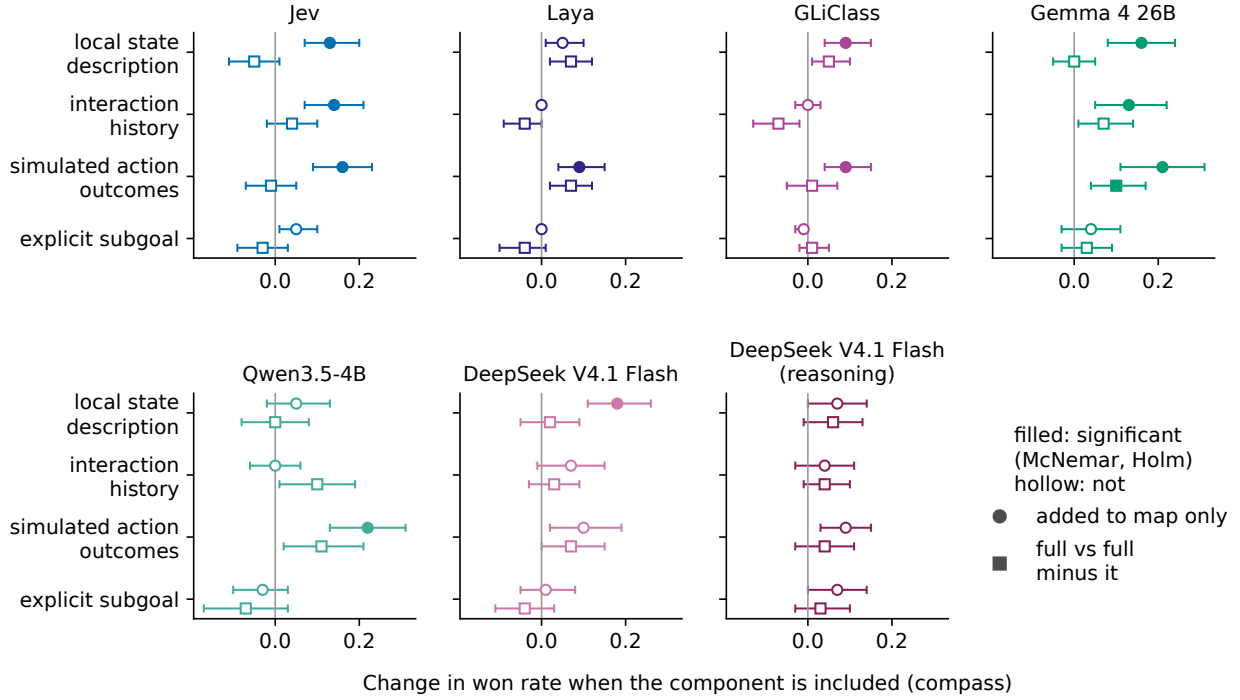


Figure 3: Paired change in won rate when a component is included. Circles compare adding it to the map alone; squares compare full context with and without it. Positive means inclusion helps. Filled markers identify McNemar tests that pass Holm correction.

adjusted won-rate test. These comparisons measure improvements over the map alone; they do not establish that simulations outperform history.

The effect depends on what else the request contains. Including history in full context adds 0.06 [0.01, 0.10] progress, whereas including simulated outcomes changes progress by 0.00 [−0.04, 0.05]. Neither corresponding won-rate contrast passes Holm correction. Our interpretation is that several descriptions may help with the same decision, so a component’s benefit alone need not mean it is needed with the others present.

5.3 Action spaces

For Jev with the map alone, won rate is 32% under *compass*, 40% under *two-moves*, 47% under *three-moves* and 26% under *up-to-three-moves*. With full context, *compass* and *up-to-three-moves* give 57% and 49%; Gemma wins 61% under both. Longer sequences do not consistently improve won rate. Option count and text length also change, preventing attribution to option count alone.

5.4 Exam

On the exam, DeepSeek V4.1 Flash (reasoning) reaches 94% [91, 96] optimality with full context. Jev and Gemma both reach 90% [87, 93]; their paired difference is 0 percentage points [−1, 2]. DeepSeek without reasoning reaches 86% [83, 90], with a Jev-minus-DeepSeek paired difference of 4 points [1, 6]. Full context improves optimality for every model: 16 points for Jev, 21 for Laya and GLiClass, 20 for Gemma, 30 for Qwen3.5-4B, 19 for DeepSeek without reasoning and 5 for DeepSeek V4.1 Flash (reasoning). All paired input-effect intervals exclude zero.

Jev’s exam optimality is 74% with the map and 90% with full context. Own-game shares pooled over moves are 16% and 64%, but averaging within puzzles and then equally over puzzles gives 38% and 79%. Long failed games dominate the pooled statistic, so part of the gap is weighting. A gap remains with equal puzzle weights: games expose model-selected positions, while the exam covers scripted positions. Jev’s full-context start and late-route scores are 86% and 97%; distance and inventory also differ, preventing isolation of history’s effect.

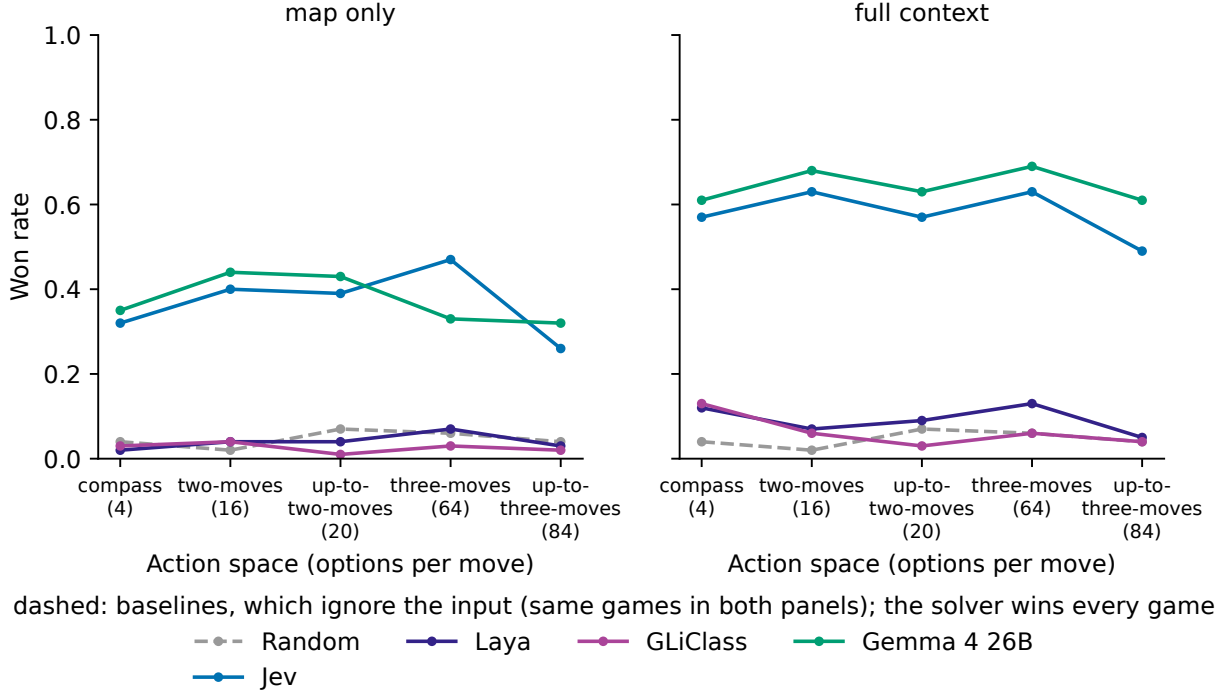


Figure 4: Won rate across compass, two-moves, up-to-two-moves, three-moves and up-to-three-moves (4, 16, 20, 64 and 84 options). Step count and observation interval also change. Qwen3.5-4B and both DeepSeek configurations ran compass moves only. Random is the same in both panels.

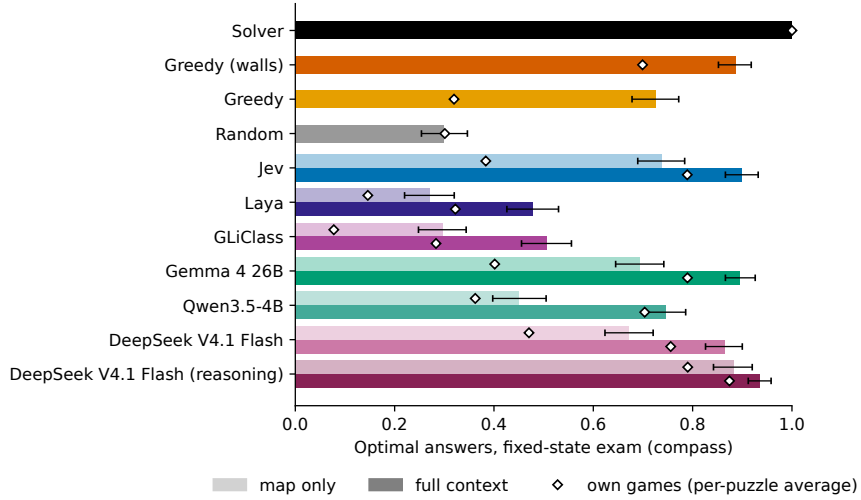


Figure 5: Optimality on the same 495 exam items. Pale and dark bars show map alone and full context; intervals resample puzzles. Diamonds show own-game optimality averaged within puzzles and then equally over puzzles, matching exam weighting. Baselines have one bar.

5.5 Blocked moves and progress

With the map alone, 82% of Jev’s moves leave the board unchanged. Without history, a blocked move leaves the board and its description unchanged, and only the option order changes, so Jev can choose the same move into the same wall again. With full context, the unchanged-board share is 0.4%, yet 43% of games still fail. Final-state progress is 0.65, below progress of 0.68. The distinction matters more for Qwen3.5-4B: its full-context scores are 0.56 and 0.66. A game can approach the goal and then finish farther away.

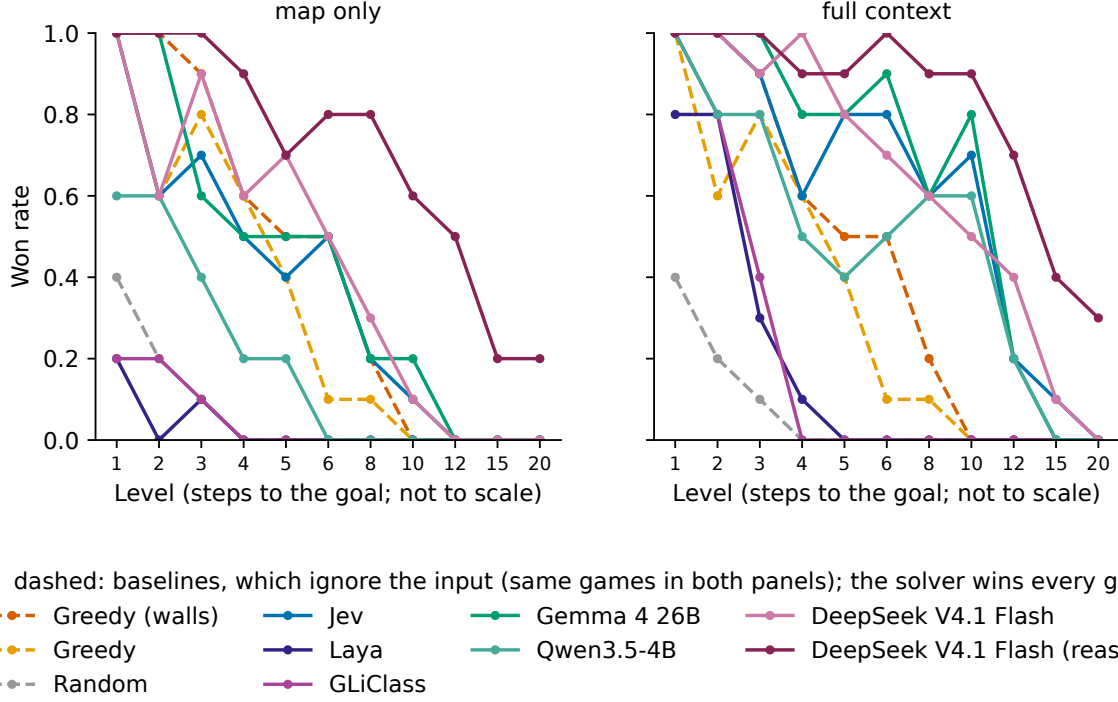


Figure 6: Compass won rate by categorical level with the map alone and full context. Labels give shortest-route distance in compass steps; horizontal gaps do not represent distance differences. Levels 1 and 2 have five puzzles each; others ten.

5.6 Won rate by level

At levels 12, 15, 20, Jev’s full-context won rates are 20%, 10%, 0%. Each level contains ten puzzles. DeepSeek V4.1 Flash (reasoning) wins 70%, 40% and 30% at those levels with full context, compared with 50%, 20% and 20% with the map alone. The higher levels remain difficult even with reasoning. These rates describe the generated levels, which vary in layout as well as shortest-route length; they do not isolate the effect of distance.

5.7 Probability analysis

Pooling compass conditions, Jev assigns its chosen move a mean probability of 0.719, while 51% of those moves are optimal. Its ECE is 0.209 and binary Brier score 0.283 over 14,642 moves. A constant at its observed optimal rate scores 0.250: an in-sample reference, not a deployable calibrated baseline. Laya’s ECE is 0.133 with 17.5% optimality; Qwen3.5-4B’s is 0.064 with 47.5% optimality. Low calibration error does not establish effective navigation. These scores concern chosen-action optimality on each model’s own positions across ten inputs. On common exam questions, Jev’s ECE is 0.063 with the map and 0.059 with full context.

TypeSafe defines Choice confidence as $c = \frac{p_{\max} - \frac{1}{n}}{1 - \frac{1}{n}}$ [17]. Across Jev’s 14,642 compass moves, the maximum discrepancy from this formula is 0.023 and remains unexplained. For a fixed option count, the formula ranks moves exactly as the top probability does; it defines a rescaled score, rather than an independent correctness prediction.

5.8 Cost and latency

These estimates pool all compass conditions. Median latency measures the successful answering call alone, excluding retries, waits and model loading: 0.503 seconds for Jev, 0.653 for Gemma, 0.805 for DeepSeek without reasoning and 6.817 for DeepSeek V4.1 Flash (reasoning). Recorded cost per 100 games is \$0.059, \$0.047 and \$0.348 for the three chat configurations, respectively. Dollar costs are unavailable for Jev and local models. Different services and hardware limit comparisons; Appendix C records the setup and accounting boundaries.

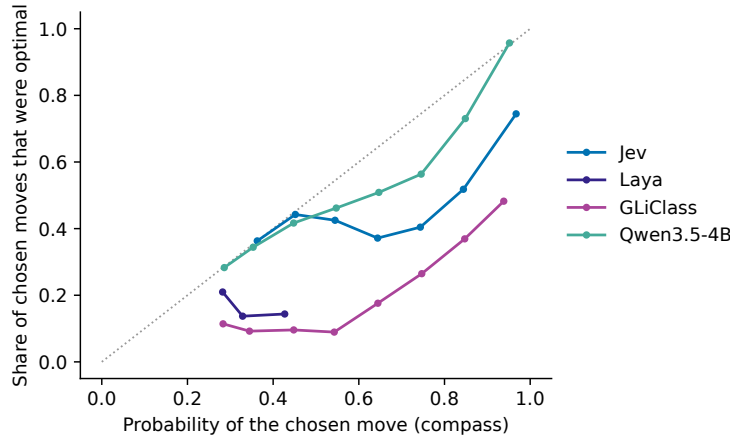


Figure 7: Reliability pooling all compass conditions; only bins with at least 30 moves are shown. Points compare mean chosen-move probability with optimality, accepting ties. The diagonal indicates agreement. Gemma and DeepSeek supply no option probabilities.

6 Discussion and limitations

Full context nearly removes Jev’s unchanged-board moves, yet 43% of games still fail. Avoiding a repeated blocked answer is only one part of navigating a route. Explicit transitions may make the map easier to use, but wording and text length are not controlled separately. Removal contrasts do not show that any component is needed with the others present, and these comparisons do not identify how the model reasons internally.

The exam/game gap remains after equal puzzle weighting: models can answer scripted individual decisions well and still fail to sustain a route. The exam samples scripted positions, while games expose the states reached by each model. Failed games also contribute repeated decisions, so pooled calibration depends on trajectories. Sequence moves change option count, text length, observation frequency and outcome duplication together; their effects cannot be isolated here.

The scope is one symbolic world, with five or ten puzzles per level and no separate held-out set. Prompts preceded puzzle generation. One trajectory per game does not measure repeated-call variation. Provider changes, inference caps and input limits can affect performance; Laya was evaluated outside its training tasks. Chat endpoints supply no option distributions. Timings use different services and hardware, and missing dollar costs limit price comparisons.

The environment, solver, frozen requests, trajectories, analysis and leaderboard are available at github.com/Exemocar/JevStuff. The repository documents submission and replay validation.

7 Conclusion

System-One Control Bench distinguishes choosing a good isolated move from completing a game. Full context improves Jev’s won rate and nearly eliminates its repeated blocked moves, yet strong exam performance does not ensure a completed route. Longer sequences have mixed effects. Jev’s moves with chosen probability $p \geq 0.9$ are more often optimal (0.74 versus 0.42 below; H7 in Table 3), although the reliability curve is not monotone. These results concern the evaluated puzzles, representations and inference settings.

AI use

This project was carried out with substantial help from AI assistants, chiefly Anthropic’s Claude models (through Claude Code) and OpenAI’s GPT models (through Codex). Under the author’s direction they wrote most of the code and the analysis scripts, drafted and revised the text of this report, and reviewed one another’s work. The author set the research question, made the design decisions, checked the results and reviewed the manuscript, and takes full responsibility for its contents.

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A One complete example request

This is the compass, full-context request for Figure 1, with every field and all state text retained. Lines wrap only where needed to fit the page. All 50 frozen requests are in the repository’s `examples/` directory.

state:

You move one cell per turn: north (up), south (down), east (right) or west (left). Walls (#) block you, and walking into one wastes the move. Stepping onto the key (K) picks it up. A locked door (D) opens when you walk into it carrying the key, and you step through; without the key it blocks you like a wall. Reaching the goal (G) wins.

Map, with column numbers along the top and row numbers down the side (# = wall, . = floor, A = you, G = goal, K = key, D = locked door). A position (x, y) is column x, row y:

```
0123456789
0 #####
1 #.....A.#
2 #...#..K.#
3 #D...#..#
4 #G#.#...##
5 #####
```

You are A at (7, 1). You are carrying nothing.

Objects: goal G at (1, 4); key K at (7, 2); locked door D at (1, 3).

North of you is a wall. South of you is the key. East of you is open floor. West of you is open floor. The goal is 6 west and 3 south of you. The key is 1 south of you. The locked door is 6 west and 2 south of you.

Moves so far:

1. east: you move to (7, 1)
2. north: blocked, you stay at (7, 1)

model: jev-1.13.0

questions.move.type: choice

questions.move.instructions: Your next target is the key K at (7, 2). Which move is the first step of the shortest path to it?

questions.move.criteria:

- option_1: move south (down): you move to (7, 2) and pick up the key
- option_2: move north (up): blocked, you stay at (7, 1)
- option_3: move west (left): you move to (6, 1)
- option_4: move east (right): you move to (8, 1)

A.1 Prompt history

Prompts were developed by inspecting a few games on one or two puzzles at a few levels before generating the benchmark puzzles. Compass text is unchanged since 4d9a877 (23 September 2026, 20:10), before Jev’s reported results in b90ab4b (22:01) and 808785f (23:48). Sequence rules entered in 4da2c7d (24 September, 00:36); their two-/three-step prompts were fixed in 5f135ed (11:43), before runs at 13:41 committed in 1144516 (14:12). Times are local, UTC+02:00. On 26 September the up-to-subgoal question was corrected to ask for the route taking the fewest turns, and every affected game was replayed by every model or baseline that had evaluated it. The analysis uses the corrected records.

B Hypotheses specified before the main evaluation

Table 3: Hypotheses specified before the main evaluation, retained verbatim. H1’s greedy comparison means no advantage demonstrated; H7’s ranking criterion is checked only by the $p \geq 0.9$ split. H2 defines its planning subset.

Hypothesis	Observed result	Criterion met
H1: under map only, Jev beats random but not greedy (walls), on won rate and progress	won vs random: +0.28 [+0.19, +0.37]; progress vs random: +0.18 [+0.09, +0.26]; won vs greedy-walls: -0.05 [-0.12, +0.02]; progress vs greedy-walls: -0.05 [-0.11, +0.01]; 'not greedy (walls)' = no advantage shown, not shown to be worse	yes
H2: Jev's wins fall with the level; no condition wins half of the planning puzzles	won-vs-level slope -0.061 to -0.046 per level; best on the 63 planning puzzles: 0.40 (everything-surroundings); planning: the 63 puzzles greedy-walls loses (100 - 37)	yes
H3: simulated action outcomes help most (best map+X, worst everything-X, on progress)	map+X: lookahead 0.61, memory 0.58, surroundings 0.57, subgoal 0.46; everything-X lowest: memory 0.63	partly
H4: explicit subgoal is second among additions, and helps mostly from level 3 up	rank 4 of 4; progress gain +0.02 at levels 3+, +0.10 at 1-2	no
H5: interaction history alone adds almost nothing (within 0.05 progress of map)	progress change +0.15 [+0.11, +0.21]	no
H6: full context has the best progress of the ten conditions and wins at least 28/100	best progress: everything-subgoal (0.73); full context wins 57/100	partly
H7: Jev is overconfident, but its p(chosen) still ranks its moves	mean p(chosen) 0.72 vs optimal share 0.51; optimal at $p \geq 0.9$: 0.74 (3895 moves), below: 0.42; ranking checked only by this split, and the reliability curve is not monotone	yes

C Reproducibility

Table 4: Evaluated model IDs and inference settings. Revisions cached on the laptop: Laya 55cf4c4, GLiClass 668dd0d, Qwen3.5-4B 851bf6e. Software dependencies are locked in `uv.lock`.

Model	Model ID / host	Precision and inference limits
Jev	jev-1.13.0 / TypeSafe	Remote precision unreported; highest option probability.
Laya	convaiinnovations/laya, subfolder typed-decisions (also published as convaiinnovations/laya-typed-decisions [6]) / local	Precision unrecorded; 8,192 request tokens, 48 per option.
GLiClass	knowledgator/gliclass-modern-large-v3.0 / local	Precision unrecorded; 8,192 tokens, up to 128 labels.
Qwen3.5-4B	Qwen/Qwen3.5-4B / local	bfloat16; thinking off; scored option numbers, no generated answer.
Gemma 4 26B	google/gemma-4-26b-a4b-it / OpenRouter, DeepInfra	Remote precision unreported; reasoning off, 64 answer tokens.
DeepSeek V4.1 Flash	deepseek/deepseek-v4.1-flash / OpenRouter, DeepInfra	Remote precision unreported; direct: 64 answer tokens; reasoning: 1,024 + 64 tokens.

The Laya and GLiClass compass games on 25 September ran on the CPU of a second computer, without a GPU; its exact model is not recorded. Every other local run (Laya and GLiClass sequence-rule games on 26 September and later, Qwen3.5-4B, and all exam runs) used an Acer Predator PHN16S-71 laptop: Intel Core Ultra 9 275HX (24 cores), 32 GB DDR5-6400, NVIDIA RTX 5070 Ti Laptop GPU (12 GB, driver 581.80), Windows 11.

Laptop software: Python 3.11.9, CUDA 13.0, torch 2.14.0+cu130, transformers 5.17.0 and gliclass 0.1.20. Runs span 23 September–3 October 2026; result filenames in `analysis/manifest.toml` identify dates. Other-machine software versions and local precision, except Qwen3.5-4B, were not recorded here.

Latency measures the successful answering call alone, excluding retries, waits and model loading. Recorded USD costs sum provider-reported answering-call costs and are scaled per 100 games; earlier failed attempts have no recorded cost. Jev and local-model dollar costs are unavailable. Host/device differences prevent a controlled hardware comparison.

D Full component effects

Each row reports both paired contrasts for one component. “Add” compares map plus the component with map alone. “Full” compares full context with full context minus that component. Positive values mean including the component helps. Intervals resample puzzles; Holm correction covers eight won-rate tests per model. Won-rate changes are proportions.

Table 5: Complete paired component effects. Brackets give unadjusted 95% intervals; p-values are Holm-adjusted. Add and Full both report the effect of including the component.

Component	Add: won change	Add: progress	Add: Holm p	Full: won change	Full: progress	Full: Holm p
Jev						
local state description	0.13 [0.07, 0.20]	0.14 [0.09, 0.20]	0.001	-0.05 [-0.11, 0.01]	-0.02 [-0.06, 0.02]	0.719
interaction history	0.14 [0.07, 0.21]	0.15 [0.11, 0.21]	<0.001	0.04 [-0.02, 0.10]	0.06 [0.01, 0.10]	1.000
simulated action outcomes	0.16 [0.09, 0.23]	0.18 [0.13, 0.24]	<0.001	-0.01 [-0.07, 0.05]	0.00 [-0.04, 0.05]	1.000
explicit subgoal	0.05 [0.01, 0.10]	0.03 [0.00, 0.07]	0.312	-0.03 [-0.09, 0.03]	-0.04 [-0.09, -0.00]	1.000
Laya						
local state description	0.05 [0.01, 0.10]	0.10 [0.06, 0.16]	0.312	0.07 [0.02, 0.12]	0.12 [0.06, 0.17]	0.109
interaction history	0.00 [0.00, 0.00]	-0.02 [-0.05, 0.00]	1.000	-0.04 [-0.09, 0.00]	0.02 [-0.03, 0.07]	0.875
simulated action outcomes	0.09 [0.04, 0.15]	0.08 [0.03, 0.14]	0.031	0.07 [0.02, 0.12]	0.09 [0.03, 0.14]	0.109
explicit subgoal	0.00 [0.00, 0.00]	-0.01 [-0.04, 0.01]	1.000	-0.04 [-0.10, 0.01]	-0.02 [-0.07, 0.02]	0.875
GLiClass						
local state description	0.09 [0.04, 0.15]	0.11 [0.05, 0.17]	0.031	0.05 [0.01, 0.10]	0.08 [0.04, 0.13]	0.312
interaction history	0.00 [-0.03, 0.03]	0.04 [0.01, 0.08]	1.000	-0.07 [-0.13, -0.02]	-0.09 [-0.13, -0.05]	0.234
simulated action outcomes	0.09 [0.04, 0.15]	0.12 [0.06, 0.18]	0.031	0.01 [-0.05, 0.07]	-0.05 [-0.12, 0.02]	1.000
explicit subgoal	-0.01 [-0.03, 0.00]	0.05 [0.01, 0.09]	1.000	0.01 [-0.02, 0.05]	0.01 [-0.02, 0.05]	1.000
Gemma 4 26B						
local state description	0.16 [0.08, 0.24]	0.18 [0.11, 0.25]	0.003	0.00 [-0.05, 0.05]	0.01 [-0.03, 0.06]	1.000
interaction history	0.13 [0.05, 0.22]	0.12 [0.06, 0.19]	0.038	0.07 [0.01, 0.14]	0.05 [0.00, 0.10]	0.262
simulated action outcomes	0.21 [0.11, 0.31]	0.22 [0.14, 0.30]	<0.001	0.10 [0.04, 0.17]	0.09 [0.04, 0.13]	0.038
explicit subgoal	0.04 [-0.03, 0.11]	0.03 [-0.03, 0.09]	1.000	0.03 [-0.03, 0.09]	-0.00 [-0.04, 0.04]	1.000
Qwen3.5-4B						
local state description	0.05 [-0.02, 0.13]	0.13 [0.06, 0.20]	1.000	0.00 [-0.08, 0.08]	0.02 [-0.03, 0.07]	1.000
interaction history	0.00 [-0.06, 0.06]	0.05 [-0.01, 0.11]	1.000	0.10 [0.01, 0.19]	0.07 [0.01, 0.14]	0.383
simulated action outcomes	0.22 [0.13, 0.31]	0.22 [0.15, 0.29]	<0.001	0.11 [0.02, 0.21]	0.10 [0.03, 0.17]	0.303
explicit subgoal	-0.03 [-0.10, 0.03]	-0.07 [-0.13, -0.01]	1.000	-0.07 [-0.17, 0.03]	-0.03 [-0.10, 0.05]	1.000
DeepSeek V4.1 Flash						
local state description	0.18 [0.11, 0.26]	0.16 [0.11, 0.23]	<0.001	0.02 [-0.05, 0.09]	-0.01 [-0.05, 0.04]	1.000
interaction history	0.07 [-0.01, 0.15]	0.09 [0.02, 0.16]	0.717	0.03 [-0.03, 0.09]	0.02 [-0.01, 0.06]	1.000
simulated action outcomes	0.10 [0.02, 0.19]	0.13 [0.06, 0.19]	0.290	0.07 [0.00, 0.15]	0.08 [0.02, 0.14]	0.711
explicit subgoal	0.01 [-0.05, 0.08]	-0.01 [-0.07, 0.04]	1.000	-0.04 [-0.11, 0.03]	-0.07 [-0.13, -0.01]	1.000
DeepSeek V4.1 Flash (reasoning)						
local state description	0.07 [0.00, 0.14]	0.05 [-0.00, 0.10]	0.646	0.06 [-0.01, 0.13]	0.03 [-0.02, 0.07]	0.730
interaction history	0.04 [-0.03, 0.11]	0.02 [-0.03, 0.08]	1.000	0.04 [-0.01, 0.10]	0.02 [-0.01, 0.06]	1.000
simulated action outcomes	0.09 [0.03, 0.15]	0.05 [0.01, 0.09]	0.094	0.04 [-0.03, 0.11]	0.02 [-0.03, 0.06]	1.000
explicit subgoal	0.07 [0.00, 0.14]	0.04 [-0.01, 0.09]	0.646	0.03 [-0.03, 0.10]	0.01 [-0.03, 0.06]	1.000